Optimal Dynamic Pricing With Preferences for Novelty

Tate Mason

The University of Georgia

November 11, 2026

Motivation

- Firms have more access than ever to consumer information
- Allows for targeted discounting based on preferences derived from repeat interactions
- Targeted discounts allow firms to steer consumers toward particular products.

How do firms price based on repeated interactions with consumers and how are profits affected?

Research Question

How can firms price discriminate via consumer preferences for novelty and habit and how much is information about these preferences worth?

How do I answer?

- Solve a theoretical model to understand firm's behavior
- Estimate consumer's preferences with Nielsen data, recovering the distribution of novelty/habit preferences
- Use the estimated distribution in the pricing model to understand how firm's revenues are impacted by this sort of targeting

Literature

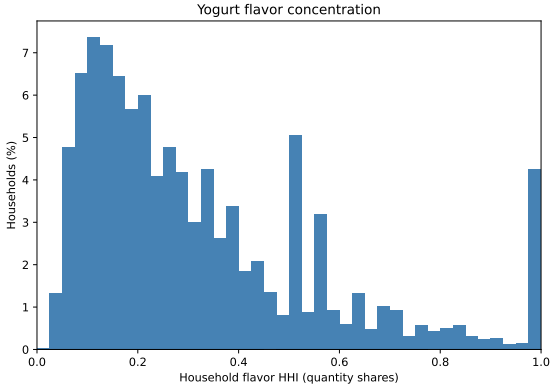
Targeting:

- Dubé (2008)
- Dubé et. al (2022)
- Anand & Shachar (2009)
- Karle & Reisinger (2024)
- Durrmeyer, Fournel, Samano (2026)

Estimating preferences for novelty:

- Dubé (2004)
- Dubé et. al (2021)
- Chintagunta (1999)
- Lattin (1987)
- McAllister (1982)
- Ackerberg (2003)

HHI



Flavor Popularity

Supply Setting

- Monopolistic firm producing a single product j
- Firm is profit maximizing, setting prices according to that objective
- A consumer is born into the world with a type $r \in \{H, L\}$, determining how much utility they derive from switching and an initial state $h_0 \in \{0, j\}$
- Consumers live for two periods, purchasing the inside good or opting for the outside good in each period
- The firm knows initial states and types, setting prices to individual consumers by type and history

Consumer's Problem

$$V_{jt}^{inside} = \underbrace{\beta}_{\text{Constant}} + \overbrace{\gamma_r \mathbb{1}\{h=0\}}^{\text{Utilty from Switching}} - \underbrace{\alpha p_t(h)}_{\text{Price Disutility}}, \quad V_{0t}^{Outside} = \underbrace{\gamma_r \mathbb{1}\{h=j\}}_{\text{Utility from Switching}}$$

$$v_t(x) = V_{jt}^{inside} - V_{0t}^{outside} = \beta + \underbrace{\gamma[\mathbb{1}\{h=0\} - \mathbb{1}\{h=j\}]}_{q(h)} - \alpha p_t(h)$$

- Prices, set by the firm, and switching utility depend on history $h = x_{t-1}$
- Utility for the outside good is normalized to zero

Firm's Problem

$$\pi_t(h) = \max_{p_t(h)} s_t(h)[p_t(h) - c]$$

$$\Pi(h) = \underbrace{s_1(h)(p_1(h) - c)}_{\text{Margin in Period 1}} + \underbrace{\delta\{s_1(h)\pi_2(j) + (1 - s_1(h))\pi_2(0)\}}_{\text{Continuation Value}}$$

- Firm profits are comprised of period one margins, given history of consumption, and the value of that consumer in the different states in period two
- The firm will set prices according to consumer type and history to maximize expected profit

First Period Optimization

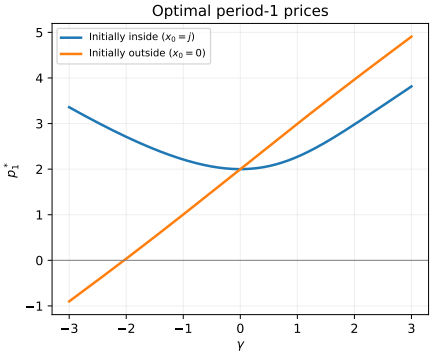
$$\frac{d\Pi}{dp_1(h_0)} = s_1 + \frac{\partial}{\partial p_1(h)} s_1(h_0)[p_1(h_0) - c + \delta(\pi_2(j) - \pi_2(0))]$$

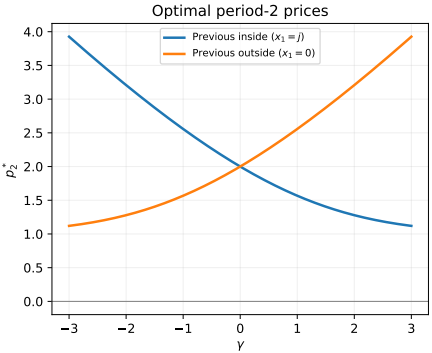
$$p_1^*(h_0) = c + \frac{1}{\alpha(1 - s_1^*(h_0))} - \delta(\pi_2(j) - \pi_2(0))$$

$$\Pi^*(h) = \delta\pi_2(0) + \frac{s_1^*(h_0)}{\alpha(1 - s_1^*(h_0))}$$

Terminal FOC Comparative statics Elasticities

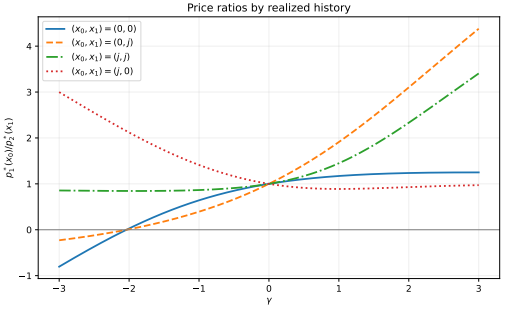
Optimal Prices

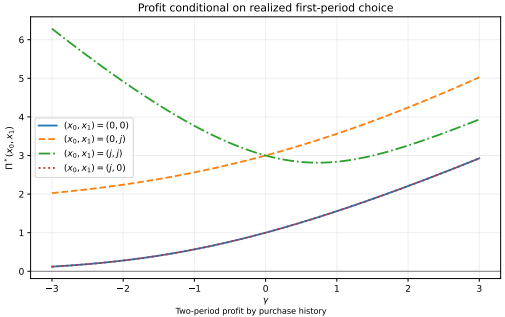




Calibration Table Profit Price ratio Discussion

Optimal Price Ratios and Profit





$\delta = 1$: no discounting. Flavor is j; outside is everything except that flavor.

▸ Optimal prices ▸ Profit detail ▸ Price-ratio detail

Two Pricing Regimes: The Same Consumer Types

Household-specific: type observed

- Observe type γ_r and initial history h_0.
- Choose $p_{1r}(h_0)$ and then $p_{2r}(h_1)$.

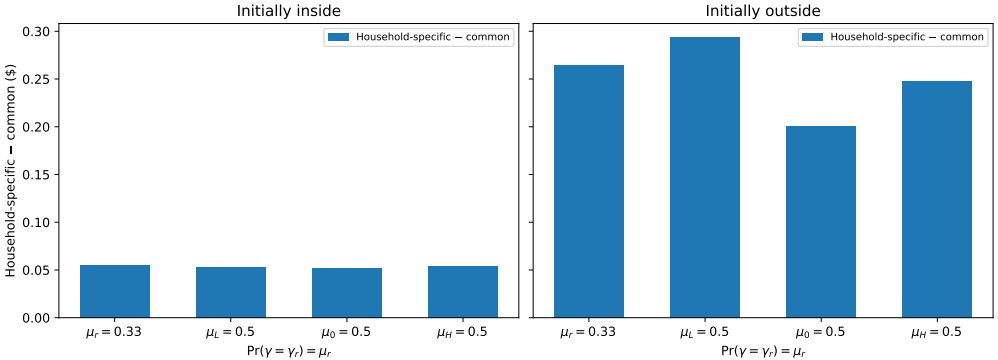
Common: type hidden

- Observe history and know type weights μ_r.
- One offer across types with the same history.
- Weights stay fixed; no Bayesian learning.

Consumers keep their type in both periods. Evaluate both policies on these same persistent types; the gain is household-specific minus common revenue.

$\gamma_r \in \{-1.25, 0, 1.25\}$; equal weights $1/3$ or permutations of $(.50, .25, .25)$. In all cases, h_0 and subsequent purchases are observed; $\delta = 1$, $c = 0$.

Theory: Revenue Gain from Observing Type



Expected revenue per consumer, evaluated under the same persistent types. Initially outside means outside the focal flavor; it is an initial state, not a population share.

Profit levels Price paths Revenue evaluation

Data Sources

Nielsen HomeScan

- Provides household level purchase data
- Covers years 2006-2015
- Products chosen, quantity chosen, timing of trips

Nielsen RetailScan

- Provides store level product data
- For specified product category: prices, markets, product attributes

Sample Description

- Must be single-person household
- Must be seen at least 2 times
- Purchases must be made at Nielsen store (prices available)
- Flavor attribute must be present

Sample Statistics: Demographics

Table: Summary statistics: full sample and yogurt sample

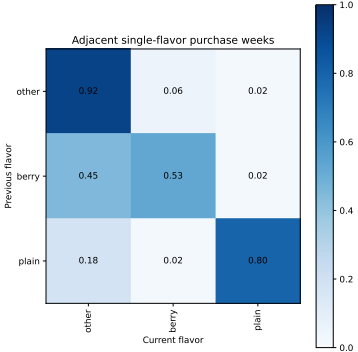
	Full sample	Yogurt sample
Sample and purchase activity		
Households	37,827	15,106
Mean observed panel years	3.96	3.16
Mean trips per approximate observed week	2.27	0.11
Mean distinct UPCs per household	1,178.31	13.89
Mean head age	45-49	50-54
Mean household income	$30,000-$35,000	$35,000-$39,999
Race: White (%)	80.33	81.58
Race: Black (%)	12.81	12.15
Race: Asian (%)	1.57	1.57
Race: Other (%)	2.19	2.11
Hispanic (%)	3.10	2.59

Switching Stats

Table: Matched yogurt sample: purchase dynamics

metric	value
Matched households	15106.00
Observed yogurt purchase weeks	282648.00
Mean flavors bought per household	1.93
Mean household flavor HHI (0 to 1)	0.78
Mixed-flavor weeks (%)	17.45
Weeks above seven packages (%)	21.72
Eligible adjacent single-flavor week pairs	48166.00
Switch rate in eligible pairs (%)	12.85
Mean single-flavor spell length (consecutive calendar weeks)	1.22
Median gap between purchase weeks	3.00

Persistence Across Aggregated Flavor Groups



Plain repeats in approximately 80% of eligible transitions, versus 53% for berry.

Estimate Heterogeneity in Price, Novelty and Habit

- Weekly choices: other yogurt, berry, plain, or nonpurchase.
- Households differ in price sensitivity and responses to recent and longer purchase histories.
- Flavor intercepts are common across households and differ across flavors.

Recover the distribution of switching preferences, then use it in the two-period pricing model.

Demand Model

$$U_{ijt} = \beta_0 + \beta_j X_{jt} + \gamma_i N_{ijt} + \phi_i H_{ijt} + \alpha_i p_{jt} + \varepsilon_{ijt}, \qquad U_{i0t} = \varepsilon_{i0t}.$$

Household i's share of yogurt purchases in flavor j in week t is

$$w_{ijt} = \frac{y_{ijt}}{\sum_k y_{ikt}}.$$

Novelty

$$N_{ijt} = \begin{cases} 1 - w_{ij,t-1}, & \text{if yogurt bought in } t-1, \\ 0, & \text{otherwise.} \end{cases}$$

Habit: fitted specification

$$H_{ijt} = \sum_{s<t} w_{ijs}.$$

Average-history specification

What Variation Separates Tastes from Purchase History?

- **Persistent tastes:** repeated choices help estimate stable differences across households.
- **Recent history:** changes in last week's basket change the novelty predictor within a household.
- **Longer history:** the same last-week basket can follow very different earlier purchase patterns.

Both histories bought only plain last week and contain 10 earlier purchase weeks. Plain's earlier share averages 80% versus 20%: novelty is identical, cumulative habit is 8 versus 2.

This describes the variation used by the model. Separating causal history effects from stable tastes also relies on the distributional, initial-history and transitory-shock assumptions.

Estimation

$$P_i(\theta) = \int P(y_i \mid \nu, \theta)\ \phi(\nu)\, d\nu$$

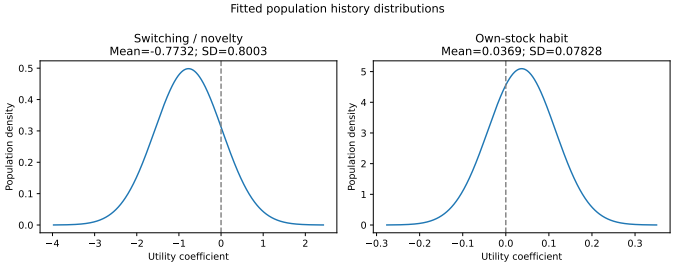
$$\widehat{P}_i(\theta) = \frac{1}{128} \sum_{r=1}^{128} \prod_{t=1}^{T_i} P(y_i \mid \nu_{ir}, \theta) \qquad \nu_{ir} \sim \mathcal{N}(0, I)$$

$$Q_{128}(\theta) = -\sum_{i=1}^{N} \log \widehat{P}_i(\theta)$$

- ν_i captures unobserved heterogreneity, y_{it} represents observed choice, $\theta = \{\beta, \gamma, \phi, \alpha\}$ is the vector of heterogeneous demand parameters
- 128 draws per household from a normal distribution
- 3 starts at different starting values, keep lowest objective
- Minimize negative log-likelihood

Probabilities Price controls Controlled estimates First stages

Distributions



▸ Estimates ▸ Discussion

Implied Willingness to Pay

Table: Implied willingness to pay without price controls

Utility term	Mean WTP	Population SD
Inside constant β_0	2.7474	2.7716
Berry vs. other	-1.4085	1.4210
Plain vs. other	-0.6137	0.6192
novelty	-1.7451	3.1116
own stock	0.0833	0.2647

- Would need to be subsidized to switch, though lots of variation
- Willing to pay to continue consuming their same product

Controlled WTP

Two Outcomes in Both Periods: Plain versus Not Plain

$$v_{irt} = b_i + \gamma_r\big[\mathbb{1}\{h_t = 0\} - \mathbb{1}\{h_t = j\}\big] + \phi_i H_{it} - a_i p_{it}.$$

- j: plain. 0: berry, other yogurt, or nonpurchase.
- Exactly one binary choice per period; the next price depends only on plain versus not plain.
- Own habit updates as $H_{i,t+1} = H_{it} + \mathbb{1}\{x_{it} = j\}$.
- Use saved estimates as inputs; fix the composite outside baseline and rival prices.

Binary extension of the theory with known habit. The outside-value mapping is an explicit counterfactual assumption, not a refit of binary demand.

Only the Firm's Knowledge of Novelty Changes

Firm knows	Household-specific prices	Common prices
Purchase history	Yes	Yes
Price sensitivity a_i	Yes	Yes
Habit preference ϕ_i	Yes	Yes
Novelty preference γ_i	Yes	Its distribution
Can respond to first purchase	Yes	Yes

Common across novelty types within a known price-sensitivity, habit and purchase-history environment.

Known preferences use estimated household means. Hidden-type weights stay fixed; consumer types persist during revenue evaluation.

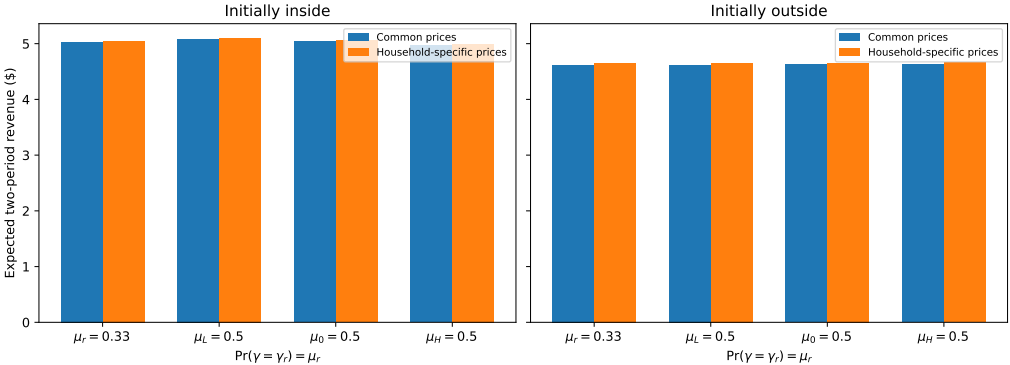
Revenue formula

Binary Plain-versus-Not Pricing: Rerun Pending

- Saved estimates are inputs to the binary model, including known habit.
- Recompute household-specific and common prices over two binary purchase periods.
- Only novelty is hidden; price sensitivity and habit are known.
- Report price changes and revenue gains in the same figure layouts as supply.

Revised results require a pricing-only server run. Earlier full-basket results do not apply to this binary specification.

Calibrated Plain Illustration: Revenue Levels



Dollars per model consumer over two periods. Both policies are evaluated under the same persistent types; $\delta = 1$.

What Does Knowing Novelty Change?

- The binary pricing model uses estimated novelty and habit; only the firm's information about novelty changes.
- Compare acquisition and repeat-purchase prices separately for initially inside and outside consumers.
- Measure the revenue gain from knowing novelty on the same persistent consumer types.

Revised pricing results pending.

Fitted-distribution baseline; expected revenue per model consumer over two periods. Both regimes adapt prices to purchase history and retain known price sensitivity and habit.

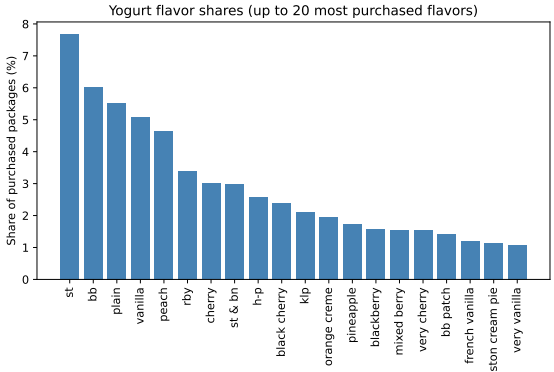
Sample Statistics: Purchase Activity

	Full sample	Yogurt sample
Households	37,827	15,106
Observed panel years	3.96	3.16
Trips per household	493.13	20.26
Trips per observed week (approx.)	2.27	0.11
Distinct UPCs per household	1,178.31	13.89
Purchase lines per household	3,397.64	45.15
Packages per household	4,669.07	93.77

All statistics except household counts are means.

Back to Data Sources

Flavor Popularity



Back to HHI

Optimal Terminal Price and Profit

$$\frac{\partial \pi_2(h_1)}{\partial p_2(h_1)} = s_2(h_1) + (p_2 - c)\frac{\partial s_2(h_1)}{\partial p_2(h_1)}$$

$$p_2^*(h_1) = c + \frac{1}{\alpha(1 - s_2^*(h_1))}$$

$$\pi_2^*(p_2^*(h_1); h_1) = \frac{s_2^*(h_1)}{\alpha(1 - s_2^*(h_1))}$$

- Optimal period two price is cost plus a demand markup term
- Period two profit at optimal prices is the odds of purchase, scaled by the inverse price sensitivity

Back to First Period Optimization

Optimal Profit Given Switching Type

$$\frac{d\Pi}{d\gamma} = \frac{q(h)s_1^*(h) - \delta s_1^*(h)s_2^*(j) + \delta(1 - s_1^*(h))s_2^*(0)}{\alpha}$$

- There are two interpretations:
  1. If $h_0 = j$, $q(h) < 0$, leading us to ambiguous signage as we raise profits from initially outside consumers but see opposing forces for the initially inside: leave in period one, return in period two
  2. If $h_0 = 0$, $q(h) > 0 \Rightarrow \frac{d\Pi}{d\gamma} > 0$, meaning that profits increase as γ increases
- Points to firms potentially reaping great benefit from knowing consumer types and histories

Back to First Period Optimization

Shares and Elasticities

$$s_t(h) = \frac{e^{v_t(h)}}{1 + e^{v_t(h)}}, \;\; s_{0t}(h) = 1 - s_t(h)$$

$$\frac{\partial s_t(h)}{\partial p_t(h)} = -\alpha s_t(h)(1 - s_t(h))$$

$$\frac{\partial s_t(h)}{\partial \gamma} = q(h) s_t(h)(1 - s_t(h))$$

Back to First Period Optimization

Pricing Simulations Specification

Parameter	Description	Value
β	Inside-good intercept	2
α	Price sensitivity	1
δ	Discount factor (no discounting)	1
c	Marginal cost	0
γ	Switching preference	$[-3, 3]$
Initial Consumption State		$h_0 \in \{0, j\}$
Feasible Prices		$p_t(h) \in \mathbb{R}$

- Allow for negative pricing
- Firm has knowledge of both type and history, allowing for targeted pricing
- Four consumption histories possible $(h_0, h_1) \in \{(0,0), (0,j), (j,j), (j,0)\}$

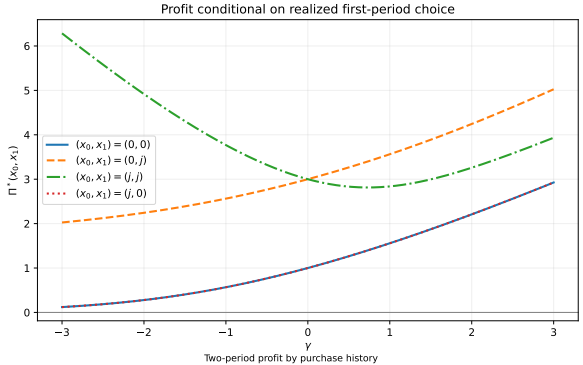
Back to Optimal Prices

Results

x_0	γ	p_1^*	$p_2^*(j)$	$p_2^*(0)$	Π^*	$\Pi^{*\prime}$
j	-3	3.358	3.926	1.120	5.284	-1.445
j	-1	2.211	2.557	1.567	2.768	-0.993
j	0	2.000	2.000	2.000	2.000	-0.500
j	1	2.271	1.567	2.557	1.838	0.177
j	3	3.814	1.120	3.926	2.934	0.730
0	-3	-0.901	3.926	1.120	1.026	0.177
0	-1	1.005	2.557	1.567	1.562	0.376
0	0	2.000	2.000	2.000	2.000	0.500
0	1	2.995	1.567	2.557	2.562	0.624
0	3	4.905	1.120	3.926	4.025	0.823

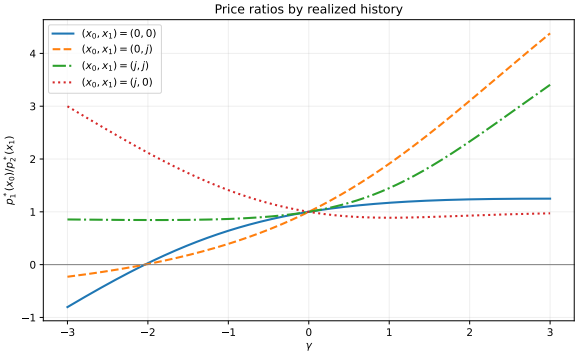
Back to Optimal Prices

Optimal Profit



Back to Optimal Prices

Optimal Price Ratio



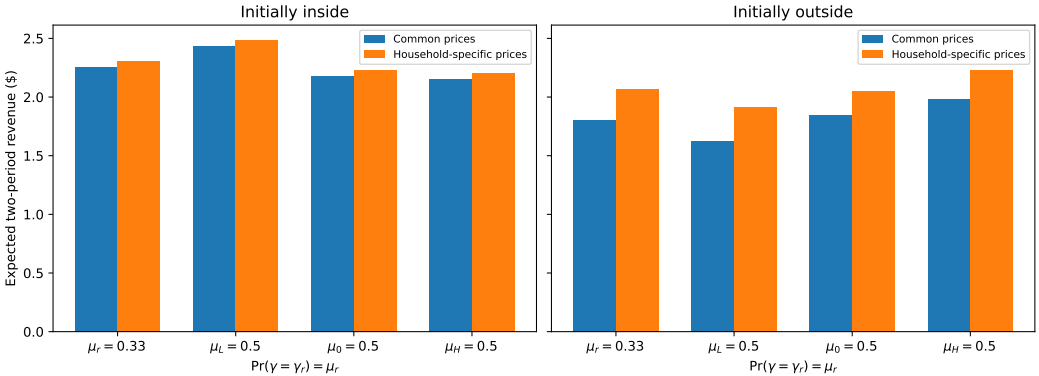
Back to Optimal Prices

Discussion

- Two mechanisms: Induce and harvest, Retain and gain
- Most profit from those who are habitual inside purchasers
- Least profit from those who purchase outside in period 1 across all levels of gamma
- Interesting when (j, j) and $(0, 0)$ cross

Back to Optimal Prices

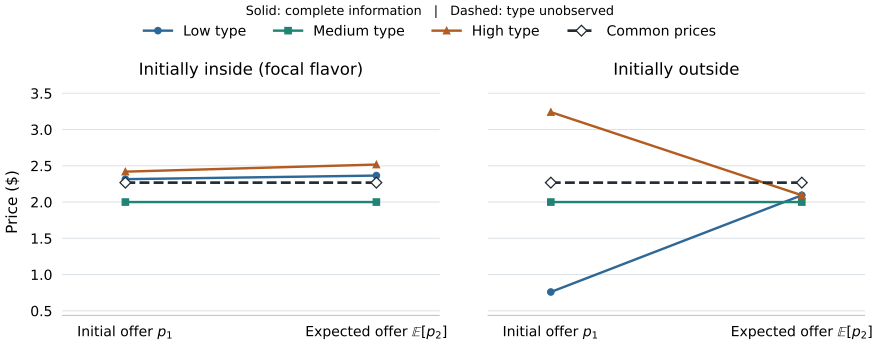
Theory: Revenue Levels under the Same Persistent Types



Full-information optimal revenue versus realized revenue of the fixed-belief common policy.

Back to Household-Specific vs Common Prices

Theory: Type-Specific and Common Offers



Equal type weights; same format as the calibrated plain price paths.

Back to Household-Specific vs Common Prices

Fixed Beliefs for Offers; Persistent Types for Evaluation

- The hidden-type firm uses μ_r in every state to choose offers; it does not update beliefs after a purchase.
- Real consumers retain their type. First-period choices sort types into histories even when the firm ignores that information.

$$\Pr(j \text{ in both periods}) = \sum_r \mu_r s_{1r} s_{2r}(j), \quad \text{not} \quad \left(\sum_r \mu_r s_{1r}\right)\left(\sum_r \mu_r s_{2r}(j)\right).$$

The firm's fixed-belief optimization objective and realized persistent-type revenue need not coincide. All displayed policy comparisons use the latter.

Back to Household-Specific vs Common Prices

Probability and Shares

$$P_{ijt} = \frac{\exp\{V_{ijt}\}}{1 + \sum_{k \in J} \exp\{V_{ikt}\}}$$

$$\text{s.t. } V_{ijt} = U_{ijt} - \varepsilon_{ijt}$$

$$s_{jt} = \frac{1}{N} \sum_{n=1}^{N} P_{ijt}$$

- Use deterministic utility to form purchase probabilities
- Take average probability for product j across individuals $i \in N$ to recover shares

Back to Estimation

Demand: Price-Control Comparison

Table: Demand estimates with and without price controls; standard errors in parentheses

Parameter	No controls	With controls
Mean novelty	-0.7732 (0.0095)	-0.7861 (0.0091)
Mean habit	0.0369 (0.0002)	0.0373 (0.0002)
SD novelty	0.8003 (0.0085)	0.7999 (0.0082)
SD habit	0.0783 (0.0004)	0.0793 (0.0004)

Back to Estimation

Controlled Demand Estimates

Table: Demand estimates with price controls

Parameter	Estimate	Standard error
Inside constant β_0	1.4762	0.0072
Berry vs. other	-0.6556	0.0034
Plain vs. other	-0.1192	0.0066
Mean price coefficient $E[\alpha_i]$	-1.0349	0.0051
Mean novelty	-0.7861	0.0091
Mean habit	0.0373	0.0002
SD price coefficient $\mathrm{SD}(\alpha_i)$	0.7957	0.0063
SD novelty	0.7999	0.0082
SD habit	0.0793	0.0004
Price control ρ	0.1959	0.0047

Back to Estimation

Controlled Fit: Price First Stages

Table: Price first stages for the controlled specification

Flavor	IV slope	Cluster SE	F	Partial R^2
Other	-34.680	0.966	1288.26	0.0213
Berry	-33.105	1.377	578.20	0.0192
Plain	-28.585	3.085	85.87	0.0076

Back to Estimation

Estimates

Table: Demand estimates without price controls

Parameter	Estimate	Standard error
Inside constant β_0	1.2173	0.0035
Berry vs. other	-0.6241	0.0034
Plain vs. other	-0.2719	0.0054
Mean price coefficient $E[\alpha_i]$	-0.8940	0.0044
Mean novelty	-0.7732	0.0095
Mean habit	0.0369	0.0002
SD price coefficient $\mathrm{SD}(\alpha_i)$	0.9019	0.0099
SD novelty	0.8003	0.0085
SD habit	0.0783	0.0004

Back to Distributions

Discussion

- Mean novelty effect is equivalent to $\approx \$1.75$ increase
- Habit provides some utility
- Both plain and berry are less preferred than other flavors

How can the firm use the distribution of switching types to price discriminate?

Back to Distributions

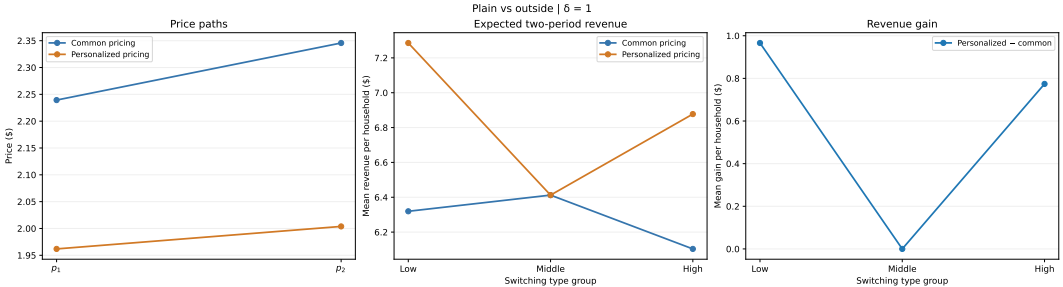
Controlled Fit: Willingness to Pay

Table: Implied willingness to pay with price controls

Utility term	Mean WTP	Population SD
Inside constant β_0	2.2696	1.7451
Berry vs. other	-1.0081	0.7751
Plain vs. other	-0.1832	0.1409
novelty	-1.2086	1.8083
own stock	0.0574	0.1600

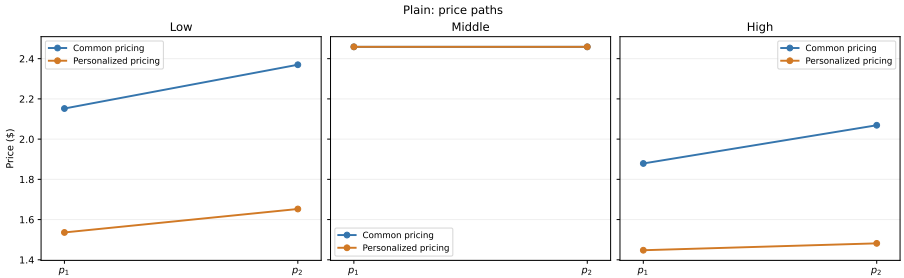
Back to Implied Willingness to Pay

Results: Plain Inside



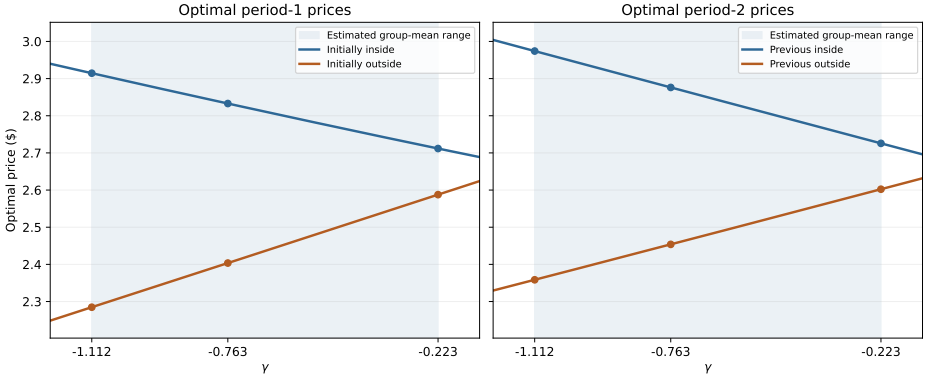
Back to Revenue Gain

Plain: Optimal Offers by Consumer Type



Back to Full-Demand Prices

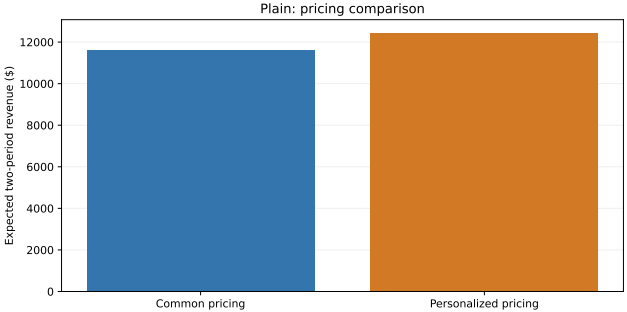
Calibrated Plain: Detail around Estimated Group Means



Same curves, zoomed around low, middle and high group means. All three means are negative. Terminal curves condition on the first-period choice.

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Separate Full-Demand Exercise: Earlier Shared-Schedule Price Benchmark



Plain: \$11,615.34 to \$12,455.27 (+7.23%) .

Same dynamic-pricing mechanism, with multinomial demand, novelty and habit. Common prices use shared percentage reductions from posted prices; household-specific prices use estimated types and basket histories. The middle group remains at posted prices. Totals are not per-consumer binary-model revenue.

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Earlier Binary Illustration: Calibration Details

- Low, middle and high groups are defined by estimated novelty preferences. Use each group's mean; the middle type is **not** set to zero.

$$(\gamma_L, \gamma_M, \gamma_H) = (-1.112,\ -0.763,\ -0.223).$$

- These are relative low/high types: all three group means are negative.
- Map novelty to the binary model's symmetric history effect, $\gamma_r[\mathbb{1}\{h = 0\} - \mathbb{1}\{h = j\}]$.
- Calibrate a common intercept and price sensitivity to observed offered prices. Keep $\delta = 1$ and $c = 0$.

Illustration, not the full split-history demand model: habit is not separately carried into this binary calibration. Price targets use 32 complete histories; treat the calibration as preliminary.

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Revenue Evaluation with Full Basket Histories

$$R(h_0) = \sum_r \mu_r \sum_{b_1 \in \mathcal{B}} P_r(b_1 \mid p_{1r}, h_0) \left[p_{1r} b_{1j} + \delta \, p_{2r}(b_1) K s_{rj}\big(p_{2r}(b_1), h_1(b_1)\big) \right].$$

- b_1 is the entire first-period binary choice; b_{1j} counts focal purchases.
- $h_1(b_1)$ updates both novelty and habit using the saved demand specification.
- Household-specific prices vary with r. Common prices do not, conditional on the other known characteristics and the same basket history.
- The same type r determines purchases in both periods.

Limited-information offers use fixed-belief optimization. Reported revenue evaluates those offers under persistent types, as in the theory comparison.

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Next Steps

- Add firm learning to supply side
- Extend market structure on supply
- Add more product attributes to empirical estimation
- Estimate unobserved quality in BLP fashion

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Average History: Planned Specification

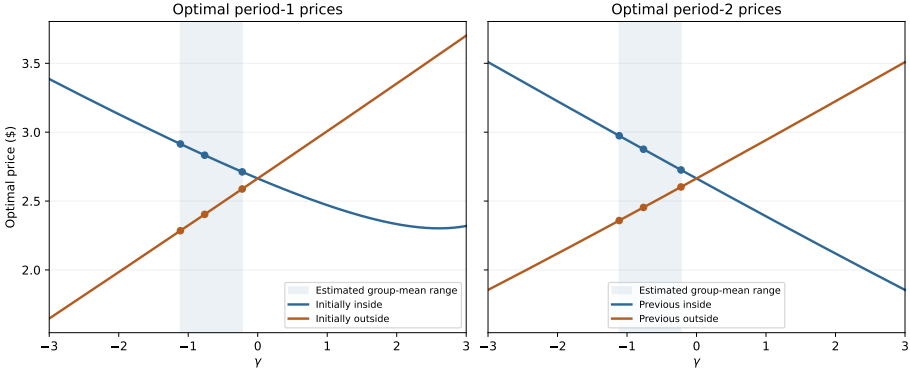
$$H_{ijt}^{\text{avg}} = \frac{1}{n_{it}} \sum_{s \in \mathcal{O}_i,\, s<t} w_{ijs}, \qquad n_{it} = \#\{s \in \mathcal{O}_i : s < t\}.$$

- $\mathcal{O}_i$: observed yogurt-purchase weeks; n_{it} counts prior occasions.
- Empty history gives zero. Unobserved calendar weeks do not enter the denominator.
- Current purchases affect the next state only.

Displayed estimates and calibrated types use the saved cumulative-history fit. Average-history estimates require a separate server refit.

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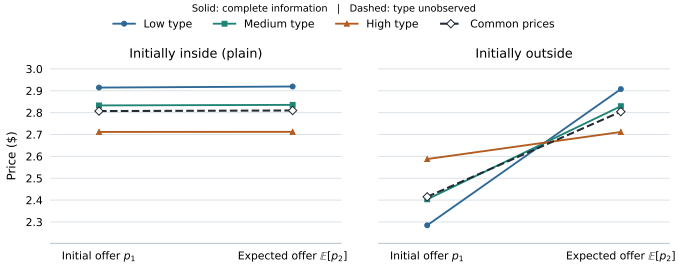
Earlier Binary Illustration: Optimal Prices across Switching Types



Same known-type model as the theory price plots; plain calibration: $\beta = 10.502$, price sensitivity $\alpha = 3.187$, $\delta = 1$, $c = 0$. Points mark estimated group means; shading spans those means. The wider γ range is a comparative-static exercise, not estimated support.

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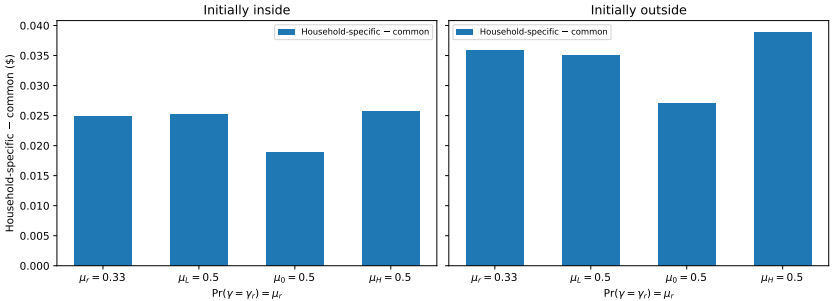
Earlier Binary Illustration: Type-Specific and Common Price Paths



Equal type weights. Paths show $p_1 \to \mathbb{E}[p_2]$. Solid paths distinguish low, middle and high types; the dashed path is the common offer. Panels distinguish initially inside and outside consumers. Terminal offers average over first-period choices.

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Earlier Binary Illustration: Revenue Gain from Observing Type



Gain per consumer over two periods, using the same persistent types under both policies.

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Earlier Binary Illustration: Information Gains by Initial State

Equal type weights	Initially inside	Initially outside
Theory: personalization gain	\$0.0555	\$0.2642
Calibrated plain: personalization gain	\$0.0249	\$0.0358

- Attracting different types can require very different offers. A common offer sacrifices the ability to tailor those incentives.
- Information value depends on the separation between optimal type-specific offers, not simply the revenue level or initial purchase rate.
- In the calibrated example, initially outside consumers generate less total revenue but a larger **incremental gain** from personalization.

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Earlier Binary Illustration: Revenue by State and Prior

Initial state	Prior	Household-specific	Common	Gain
j	Equal types	5.0508	5.0259	0.0249
0	Equal types	4.6564	4.6205	0.0358
j	Low type more likely	5.1033	5.0781	0.0252
0	Low type more likely	4.6501	4.6151	0.0350
j	Medium type more likely	5.0584	5.0396	0.0189
0	Medium type more likely	4.6553	4.6282	0.0271
j	High type more likely	4.9906	4.9649	0.0257
0	High type more likely	4.6637	4.6248	0.0389

Dollars per model consumer, evaluated under persistent types. j: initially plain; 0: initially outside plain, including other yogurt. Price targets use 32 complete histories; this is an illustrative calibration.

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